COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
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ACTIVITY NO. 11 **Containerization**

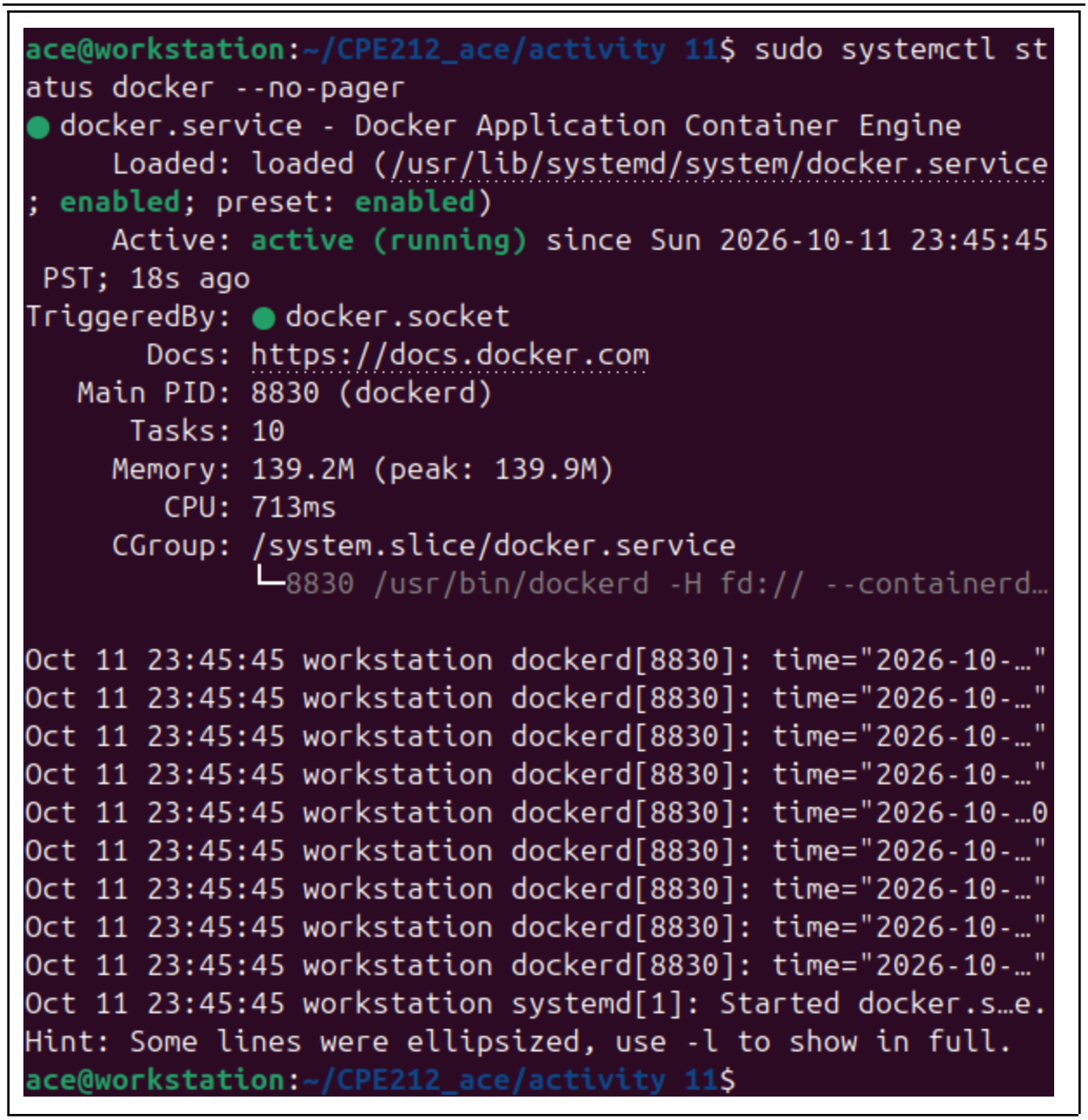
1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

```
ace@workstation:~$ ls
ansible-lab  Documents  Pictures  Templates
CPE212_ace   Downloads  Public    Videos
Desktop      Music      snap
```

2. Install Docker and enable the docker socket.

```
ace@workstation:~/CPE212_ace/activity 11$ sudo apt install
docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```


ace@workstation:~/CPE212_ace/activity 11$ sudo systemctl st
atus docker --no-pager
● docker.service - Docker Application Container Engine
Loaded: loaded (/usr/lib/systemd/system/docker.service
; enabled; preset: enabled)
Active: active (running) since Sun 2026-10-11 23:45:45
PST; 18s ago
TriggeredBy: ● docker.socket
Docs: https://docs.docker.com
Main PID: 8830 (dockerd)
Tasks: 10
Memory: 139.2M (peak: 139.9M)
CPU: 713ms
CGroup: /system.slice/docker.service
└8830 /usr/bin/dockerd -H fd:// --containerd…
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…0
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation dockerd[8830]: time="2026-10-…"
Oct 11 23:45:45 workstation systemd[1]: Started docker.s…e.
Hint: Some lines were ellipsized, use -l to show in full.
ace@workstation:~/CPE212_ace/activity 11$

```
ace@workstation:~/CPE212_ace/activity 11$ sudo docker run h
ello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df96
05b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be wor
king correctly.
```

3. Add to Docker group to your current user.

```
ace@workstation:~/CPE212_ace/activity 11$ sudo groupadd -f
docker
[sudo] password for ace:
ace@workstation:~/CPE212_ace/activity 11$ sudo usermod -aG
docker "$USER"
ace@workstation:~/CPE212_ace/activity 11$ newgrp docker
```

4. Create a Dockerfile to install web and DB server.

```
ace@workstation:~/CPE212_ace/activity 11$ nano Dockerfile
ace@workstation:~/CPE212_ace/activity 11$ cat Dockerfile
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y apache2 mariadb-server && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD service mariadb start && apache2ctl -D FOREGROUND
```

5. Install and build the Dockerfile using Ansible.

```
ace@workstation:~/CPE212_ace/activity 11$ ansible-playbook
-i inventory.ini build_docker.yml

PLAY [Build the web and database Docker image locally] ****
**********************

TASK [Gathering Facts] ************************************
**********************
ok: [localhost]

TASK [Create the Docker build directory] ******************
**********************
changed: [localhost]

TASK [Copy the Dockerfile] ********************************
**********************
changed: [localhost]

TASK [Build the web and database image] *******************
**********************
```

```
TASK [Build the web and database image] ********************
**********************
changed: [localhost]

TASK [List Docker images] ***********************************
**********************
ok: [localhost]

TASK [Display Docker images] ********************************
**********************
ok: [localhost] => {
    "docker_images.stdout_lines": [
        "IMAGE                  ID             DISK USAGE
 CONTENT SIZE   EXTRA",
        "hello-world:latest     5e2309035332       25.9kB
       9.49kB   U    ",
        "web-db-server:latest   eddd512eb3de        616MB
        116MB        "
    ]
}

PLAY RECAP **************************************************
**********************
localhost                  : ok=6    changed=3    unreachab
le=0    failed=0    skipped=0    rescued=0    ignored=0
```

6. Add, commit and push it to your repository.

```
ace@workstation:~/CPE212_ace$ git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 3 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (7/7), 872 bytes | 436.00 KiB/s, done
.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 loca
l objects.
To github.com:Ace-CpE/CPE212_ace.git
   c1cd2ae..920d744  main -> main
ace@workstation:~/CPE212_ace$
```

Answer the following:
1. What are the benefits of implementing containerizations?

Containerization provides resource efficiency, consistency, faster deployment, and easier to manage. It allows applications to run reliably across different existing environments at the same time it uses much fewer resources than traditional virtual machines.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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